

Appendix F: Ethics, Reproducibility and Final Audit Material

F.1 Purpose

This appendix records ethics, data-handling, reproducibility, risk analysis and audit material for the dissertation. It also retains pilot and pre-main-run validation evidence as method-validation and audit material only, so that preparation work is not confused with the final main-run findings reported through Appendix E and Chapter 5.

F.2 Ethics and Data-Handling Statement

The experimental corpus consists of publicly available SEC Form 10-K disclosures. The methodology does not intentionally collect private personal data, confidential corporate intellectual property, proprietary analyst research, personal financial information or non-public corporate records.

Model-produced labels and rationales are experimental artefacts within a fixed-condition computer-science research framework. They must not be interpreted as investment advice, audit conclusions, legal determinations, financial recommendations or judgements about the underlying companies.

API keys, credentials, local environment files and other secrets are excluded from submitted artefacts. Source filings, prompts, outputs, telemetry, scoring records and audit evidence are retained only for dissertation reproducibility and examination of the experimental method.

F.3 Pilot and Pre-Main-Run Validation Boundary

Pilot and pre-main-run materials are retained to document method calibration, apparatus validation and risk control. The pilot used separate pilot/stress claim-cases and is not used as Chapter 5 final comparative evidence. Pre-main-run checks validated the final apparatus and locked inputs before the controlled main run. These materials do not alter the locked main-run scoring key, final claim-case manifest or final result evidence reported in Appendix E.

The pre-main-run reports and source ZIPs retained under `05_Audit_and_Control/` are historical audit artefacts created before the final 96-call main execution. Any references in those files to the main experiment as “not started” describe the status at the time those validation artefacts were prepared, not the final dissertation status. They are retained only to evidence validation, prompt dry-run, smoke testing, stress testing, RuntimePatch001 and readiness controls before the main run.

Table F.1: Pilot and pre-main-run validation boundary

Pilot stage	Scope	Outcome	Methodological interpretation
-------------	-------	---------	-------------------------------

Pilot stage	Scope	Outcome	Methodological interpretation
Tiny pilot	4 model calls: two Apple claim-cases under both context-delivery conditions.	4 of 4 returned the expected label with valid JSON.	API execution, schema adherence and base logging validated.
JPMorgan stress micro-check	2 model calls: one JPMorgan claim-case under both context-delivery conditions.	2 of 2 returned the expected label with valid JSON.	Large single-filing long-context path and comparable RAG path validated.
Full pilot subset	18 model calls: nine claim-cases under both context-delivery conditions.	15 of 18 returned the expected label.	Unsupported-causality and RAG retrieval-completeness risks were exposed before the main run.
Pre-lock calibration check	3 affected model calls.	3 of 3 targeted calls returned the expected label after calibrated rules.	Unsupported-causality rules and retrieval-completeness checks under the fixed overlapping-word-window configuration were stabilised before the locked main run.

F.4 Calibration and Apparatus Consolidation Before Main Run

The dissertation apparatus was developed iteratively. Earlier scripts were used for pilot validation, SEC filing smoke tests, clean-text preparation, claim-case construction support, RAG-context preparation and prompt dry-run checks.

After the final six-filing corpus, 48 claim-case manifest, scoring key, prompt set, RAG contexts and execution queue were locked, the workflow was consolidated into the containerized apparatus used for final validation, prompt generation, live model execution, telemetry capture, scoring and evidence export. This separation preserves the audit trail between pilot/calibration work, pre-main-run validation checks and the controlled final main experiment.

Table F.2: Calibration and apparatus consolidation decisions

Calibration issue	Locked decision
Unsupported causal claims	Default to Insufficient Evidence unless the supplied filing evidence directly supports or directly contradicts the causal relationship.
RAG retrieval completeness	For tabular, segment-level or distributed-evidence claim-cases, retrieved context must contain enough filing evidence for a fair RAG decision; retrieval miss remains a RAG limitation rather than a post-hoc correction point.
Runtime request alignment	RuntimePatch001 records that explicit temperature control was omitted after the selected GPT-5.5 API execution path rejected an explicit temperature parameter. The same final request path was used for both context-delivery conditions.
Main-run guardrail	After parameter, prompt, RAG, label and scoring lock, no calibration may be applied based on observed main-run outputs.

F.5 Reproducibility Checklist

Table F.3: Reproducibility checklist

Reproducibility item	Audit evidence and submitted location
Appendix A literature and citation-control material retained	Appendix A files are retained under 02_Appendices/ and record the literature protocol and citation-control matrix.
Appendix B parameter-lock and execution protocol retained	Appendix B is retained under 02_Appendices/ and records the experiment protocol, parameter lock, retry rules, logging fields and cost controls.
Appendix C filing register and claim-case manifest retained	Appendix C is retained under 02_Appendices/ and remains the source of truth for filing selection, claim-case text, expected labels, anchors, rationales and scoring rubric.
Appendix D prompt, schema and RAG configuration retained	Appendix D is retained under 02_Appendices/ and records executed prompt templates, shared output schema and locked RAG configuration.
Appendix E final main-run evidence retained	Appendix E is retained under 02_Appendices/; supporting evidence is retained under 03_Main_Run_Evidence/.
Pilot/calibration evidence retained separately	Pilot validation summary and evidence workbook are retained under 05_Audit_and_Control/; they are method-validation material only and are not Chapter 5 final result evidence.
Pre-main-run validation evidence retained separately	Pre-main-run Markdown reports and supporting source ZIPs are retained under 05_Audit_and_Control/Pre_Main_Run_Reports/ and 05_Audit_and_Control/Pre_Main_Run_Source_Zips/. They cover apparatus validation, prompt dry-run, smoke-live, runtime-patch, largest-prompt stress-smoke and full-live readiness evidence.
Raw and parsed final main-run outputs retained	Raw and parsed final main-run outputs are retained in 03_Main_Run_Evidence/Locked_Source_Zips/main_run_full_live_outputs_locked_v1.zip.
Cost, token and latency records retained	Telemetry records are retained in 03_Main_Run_Evidence/Appendix_E_Evidence_Pack/ and in the derived analysis workbook under 03_Main_Run_Evidence/Derived_Analysis/.
Final software and environment artefacts retained	The final containerized codebase and reproducibility environment/configuration files are retained under 04_Codebase/CS958-ROUS04-Codebase.zip; API keys, credentials and local secret files are excluded.
No API keys or local secrets submitted	Verified: credentials, API keys, local .env files and private runtime settings are excluded from the submitted dissertation artefact package.

F.6 Risk and Mitigation Register

This register records the main methodological and operational risks for the controlled main experiment. It is not used to change the experiment after results are known; final empirical interpretation remains in Chapters 5 and 6.

Table F.4: Risk and mitigation register

Risk	Impact	Likelihood	Mitigation / locked control
Long-context API cost or latency escalation	High	Medium	Use locked 6-filing / 48-claim-case scale, dry-run checks, token/cost telemetry, USD 125 hard ceiling and USD 90 warning threshold.
API rate limit, timeout or temporary provider unavailability	Medium	Medium	Apply predefined transient retry rules, log timestamps/error codes and avoid unlogged reruns or manual context-delivery-condition-specific correction.
Hidden model-provider update or model drift	Medium	Low	Record model identifier, provider-returned metadata where available, run date/time and paired execution order.
SEC 10-K extraction or section-coverage error	High	Medium	Validate cleaned text, section coverage, token estimates, accession metadata, report dates and smoke-test status before main execution.
RAG retrieval miss, retrieval noise or chunk-boundary fragmentation	High	Medium	Freeze chunk size, overlap and top-k; retain retrieved chunk identifiers, ranks, scores, word offsets and text for post-run diagnosis.
Malformed JSON, schema failure or unsupported output format	Medium	Medium	Use strict JSON-schema enforcement, predefined parse/invalid-output handling, and retained raw and parsed output records.
Ground-truth label or qualitative error-coding subjectivity	Medium	Medium	Use documented filing-reference anchors, label-decision rules, expected-label rationales and exclusion of unresolved ambiguous claim-cases.
Answer-key leakage into model-facing calls	High	Low	Keep expected labels, ground-truth rationales and target evidence snippets outside model-facing calls; validate leakage controls before execution.
Confusion between pilot evidence and final main-run findings	Medium	Medium	Retain pilot/pre-main-run evidence in Appendix F and final main-run evidence in Appendix E.
Secret leakage in submitted artefacts	High	Low	Exclude API keys, environment files, local credentials and private runtime settings from the final submission package.
Uncalibrated model-reported confidence	Medium	Medium	Retain confidence descriptively only; do not use it as a reliability gate. All 13 incorrect main-run outputs reported High confidence.

F.7 Artefact Register and Audit Location

The register below identifies the main submitted artefact groups and their audit role. Paths are expressed relative to the submitted dissertation artefact package, `Dissertation_Submission_Artefacts.zip`, where applicable.

Table F.5: Artefact register and audit location

Artefact / location	Audit role and boundary
---------------------	-------------------------

Artefact / location	Audit role and boundary
01_Report/Dissertation_Final_Report.pdf	Final report containing Chapters 1 to 6. Uses appendices as supporting evidence rather than embedding full run files in the report body.
02_Appendices/Appendices_A_to_F_Pack_Final.zip	Final appendix pack containing the readable supporting appendices A to F.
02_Appendices/Appendix_A1_* and Appendix_A2_* files	Literature-review protocol and citation-control matrix; controls source use and citation boundaries.
02_Appendices/Appendix_B_Methodology_and_Experiment_Protocol_Final.pdf	Operational protocol for the controlled experiment; records parameter lock, execution rules, retry handling, logging and cost controls.
02_Appendices/Appendix_C_Dataset_and_Claim_Case_Register_Final.pdf	Dataset and locked claim-case register; source of truth for claim-case text, labels, anchors and rationales.
02_Appendices/Appendix_D_Prompt_Templates_Output_Schema_and_RAG_Configuration_Final.pdf	Executed prompt templates, schema and RAG configuration; source for the model-facing task and context-delivery configuration.
02_Appendices/Appendix_E_Experiment_Run_Logs_Cost_Latency_and_Results_Evidence_Final.pdf	Final main-run evidence appendix; summarises validation, telemetry, scoring, metrics, confusion matrices and qualitative error evidence.
02_Appendices/Appendix_F_Ethics_Reproducibility_and_Audit_Material_Final.pdf	Ethics, reproducibility and final audit appendix; records public-data handling, pilot/pre-main-run validation boundary, reproducibility controls, risk controls and audit artefact locations.
03_Main_Run_Evidence/Appendix_E_Evidence_Pack/	Curated Appendix E evidence pack; primary Appendix E evidence route for final main-run review.
03_Main_Run_Evidence/Derived_Analysis/Main_Run_Results_Extraction_and_Analysis_Workbook_v1.xlsx	Post-run derived analysis baseline; immediate source for Appendix E summary tables derived from locked main-run evidence packages.
03_Main_Run_Evidence/Locked_Source_Zips/main_run_full_live_outputs_locked_v1.zip	Locked full-live source package; source evidence for model-call payloads, raw/parsed outputs, prompt hashes and telemetry.
03_Main_Run_Evidence/Locked_Source_Zips/main_run_scoring_outputs_locked_v1.zip	Locked scoring source package; source evidence for scored predictions, condition metrics and confusion matrices.

Artefact / location	Audit role and boundary
03_Main_Run_Evidence/Locked_Source_Zips/appendix_e_evidence_pack_locked_v1.zip	Runtime-exported Appendix E evidence package; curated export produced by the evidence-export workflow and not a replacement for full-live or scoring source packages.
04_Codebase/CS958-R0US04-Codebase.zip	Final containerized codebase and reproducibility environment; supports validation, prompt generation, live execution, telemetry capture, scoring and evidence export; excludes API keys, credentials and local secret files.
05_Audit_and_Control/Pilot_Validation_Summary_for_Chapter_3_Methodology.pdf 05_Audit_and_Control/Pilot_Validation_Evidence_Workbook.xlsx	Pilot validation summary and evidence workbook; audit-only support for methodology calibration and not final Chapter 5 evidence.
05_Audit_and_Control/Pre_Main_Run_Reports/	Pre-main-run Markdown reports manually prepared from retained apparatus validation, prompt dry-run, smoke-live, runtime-patch, largest-prompt stress-smoke and full-live readiness evidence. Audit summaries only; not final main-run result evidence.
05_Audit_and_Control/Pre_Main_Run_Source_Zips/	Locked pre-main-run source ZIPs containing validation/prompt-dry-run outputs, paired smoke-live artefacts, largest-prompt stress-smoke artefacts and RuntimePatch001 core files. These packages retain the underlying artefacts for audit and are not Chapter 5 result evidence.

F.8 Final Audit Boundary

The controlled main experiment was executed after the filing corpus, claim-case manifest, scoring key, prompts, schema and RAG configuration had been locked. Final execution and scoring evidence is reported through Appendix E; pilot and pre-main-run evidence in this appendix is retained only to demonstrate how the apparatus and methodological safeguards were validated before the final run.

RuntimePatch001 (05_Audit_and_Control/Pre_Main_Run_Source_Zips/main_run_container_app_v1_runtimepatch001_core_files.zip) is treated as a runtime-alignment limitation rather than a change to the independent variable, because the same final request path was applied to both context-delivery conditions. Exact output determinism is not claimed. The dissertation findings should therefore be interpreted as a controlled same-model comparison over the locked six-filing, 48-claim-case dataset, not as a universal benchmark for all financial filings or all model deployments.

The audit boundary also records that one execution was performed per condition-case pair; no independent dual-coding or inter-rater agreement exercise was conducted; confidence was not calibrated; no positional, attention or lost-in-the-middle analysis was performed; and concurrency, throughput and

production-scale load were not tested. Reproducibility is therefore supported through locked inputs, consistent procedures and retained evidence rather than a claim of exact output determinism under future provider-side changes.

No further model calls, reruns, claim-case changes, scoring-key changes or post-hoc retrieval repairs are part of the final main-run evidence boundary.